Ahmed Muharram

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SUMMARY

Software engineer working on LLM agents, search systems, and cloud infrastructure. Took Fidelity's internal RAG search to production for about 200 daily users and built PathFinder, an open-source agent for VEuPathDB's 14 genomics databases.
Egyptian citizen. Military service: Exempt.

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science — Philadelphia, PA
M.S.E. in Computer Science, GPA 3.84 — *Apr 2024 – May 2026*
B.S.E. in Computer Science, GPA 3.77 — *Aug 2022 – May 2026*
Penn Engineering Outstanding Research Award (2026). Onsi Sawiris Scholar, a full scholarship for Egyptian students at US universities. Minors in Mathematics, Data Science, and Engineering Entrepreneurship.

EXPERIENCE

Software Engineer (IT Applications Consultant), Orascom Construction, R&D — *Cairo, Egypt | Jul 2026 – Present*

- Rewriting the company's 2012 ASP.NET IT request and approval system in .NET 10, Angular, and PostgreSQL as the only engineer, now in user acceptance testing.
- Migrated 40K+ requests, 200K+ comments, and 10+ GB of attachments from SQL Server to PostgreSQL.
- Built and administer the AWS platform and GitHub Actions deploy pipelines the R&D group's 4 apps run on (ECS on Graviton, RDS, CloudFront, WAF, CloudFormation), with separate UAT and production clusters.

Fidelity Investments — *Jun 2025 – Jan 2026*
AI/ML Engineer — *Jersey City, NJ | Dec 2025 – Jan 2026*

- Took the internal RAG search system from prototype to production, where about 200 R&D staff use it daily.
- Cut index refresh time from about 10 minutes per million chunks to under a second by moving retrieval from BM25S to PostgreSQL full-text search and pgvector.
- Added late-interaction reranking, sharded the Qdrant cluster, and backfilled historical embeddings into pgvector.
- Built a connector and ingestion template for adding new enterprise data sources.

Data Science and AI Intern — *Smithfield, RI | Jun 2025 – Aug 2025*

- Built the first version of the RAG system on Qdrant and pgvector, including an ingestion pipeline that processed terabytes of internal documents at a 99.99%+ success rate.
- Added grounding checks and a retrieval evaluation harness, and deployed the services on Docker and Kubernetes.

Software Engineering Intern (AI Integration), Aydi — *Maadi, Cairo | Jun 2024 – Aug 2024*

- Built an analytics chatbot for farm businesses (Haystack, MongoDB, Redis) with 85%+ top-1 retrieval accuracy.
- Cut the chatbot's LLM costs 96.25% by switching from GPT-4o to GPT-4o mini after cost and quality benchmarking.

PROJECTS

PathFinder, open-source research agent for VEuPathDB — github.com/ahmedomuharram/pathfinder

- Designed and built, as the sole developer, an agent that turns researchers' questions into multi-step searches across VEuPathDB's 14 pathogen-genomics databases, now running on VEuPathDB's QA servers.
- Structured it as a lead agent with planning and verification sub-agents (pydantic-ai, LangGraph, 44 tools) that compiles each question into a strategy graph of WDK searches, Boolean operators, and ortholog transforms, which researchers can edit in a ReactFlow editor.
- Replaced the LLM build step with deterministic code, grounded the agents in each site's live catalog with pgvector semantic search, and moved agent turns onto background workers that stream over SSE.
- Extracted 3 libraries into the VEuPathDB GitHub organization: an MCP server for the site catalog, an assistant runtime, and a typed API client.

LAMINA, materials-feasibility agent github.com/ahmedomuharram/lamina

- Built a 50-tool agent that checks materials-science feasibility claims with CALPHAD phase diagrams, Materials Project DFT data, and CHGNet neural surrogates, evaluated on 37 expert-labeled DARPA SciFy Drop-4 claims.

ScenIQ, on-prem meeting intelligence *Penn senior design project*

- Co-built an air-gapped pipeline that turns meeting recordings into transcripts and assigned action items with FFmpeg, speaker diarization, and Whisper, processing a 5-minute video in 1 to 2 minutes.

Other projects

- ByteNet, a distributed search engine in Java on AWS EC2 with a crawler that fetched about 500K pages, a TF-IDF index, and PageRank.
- InstaLite, an Instagram clone on AWS with semantic profile search and real-time chat over WebSockets.
- An RV32IM RISC-V processor in SystemVerilog with a pipelined datapath, verified with Verilator.
- PennOS, a UNIX-like operating system simulator in C with a preemptive priority scheduler and a FAT-style file system.
- Mini-Minecraft, a voxel sandbox in C++ and OpenGL with chunked terrain rendering.

PUBLICATIONS

D. Rao, **A. Muharram**, C. Callison-Burch. Same Claim, Different Interpretations: How LLM Agents Resolve Ambiguity in Scientific Claims. *arXiv preprint*, 2026. arxiv.org/abs/2610.20457

- Found that changing one field of an agent's written interpretation flipped its verdict on about a third of 60 expert-labeled claims across 900 GPT-5.4 runs, compared with about one in seven by chance ($p < 0.001$).

A. Muharram. How Underspecified Prompts Shape Tool-Calling LLM Agents in Scientific Workflows. *Master's thesis, University of Pennsylvania*, 2026. Advised by Chris Callison-Burch. muharram.dev/thesis

- Showed across two agents (LAMINA and PathFinder) and five OpenAI model setups that prompt vagueness caused more agent errors than model size, routing, or task difficulty.
- In a 420-run study on a 35-strategy gold set, PathFinder's gene-set F1 fell from 0.66 to 0.23 to 0.16 as prompts went from precise to vague. LAMINA's verdict flip rate on vague claims was 0.43.
- Found that the errors came from how agents turned prompts into tool calls, while tool execution itself stayed reliable.

SKILLS

AI and LLMs: agentic AI, agent harnesses, tool-calling and multi-agent systems, RAG, MCP, evals, prompt and context engineering; pydantic-ai, LangGraph, LangChain, Haystack, Vercel AI SDK; OpenAI, Anthropic, and Gemini APIs; AI-assisted development (Claude Code)

Retrieval and Data: pgvector (HNSW), Qdrant, PostgreSQL full-text search, hybrid and late-interaction retrieval, reranking; PostgreSQL, SQL Server, Redis, MongoDB

Languages and Frameworks: Python, C#, TypeScript, SQL, Java; FastAPI, ASP.NET Core, EF Core, Angular, React

Cloud and DevOps: AWS (ECS, RDS, S3, CloudFront, WAF, CloudFormation, SSM, SQS, SES), Azure AI Foundry, Docker, Kubernetes, GitHub Actions, OIDC, OpenTelemetry, Langfuse

Languages Spoken: Arabic and English (native); limited German and Italian

CERTIFICATIONS

Microsoft Certified: Azure AI Apps and Agents Developer Associate (2026)